

Creation Date 29-Oct-2010

Revision Date 25-Sep-2023

Revision Number 9

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                                   |
|----------------------------------|---------------------------------------------------|
| <b>Product Description:</b>      | <b>Isobutyric acid</b>                            |
| <b>Cat No. :</b>                 | <b>122520000; 122520010; 122520250; 122525000</b> |
| <b>Synonyms</b>                  | 2-Methylpropionic acid                            |
| <b>Index No</b>                  | 607-063-00-9                                      |
| <b>CAS No</b>                    | 79-31-2                                           |
| <b>EC No</b>                     | 201-195-7                                         |
| <b>Molecular Formula</b>         | C4 H8 O2                                          |
| <b>REACH registration number</b> | 01-2119488973-18                                  |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Flammable liquids

Category 3 (H226)

##### Health hazards

Acute oral toxicity

Category 4 (H302)

Acute dermal toxicity

Category 3 (H311)

Skin Corrosion/Irritation

Category 1 B (H314)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

Danger

#### **Hazard Statements**

H226 - Flammable liquid and vapor

H302 - Harmful if swallowed

H311 - Toxic in contact with skin

H314 - Causes severe skin burns and eye damage

#### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

### 2.3. Other hazards

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Stench

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component       | CAS No  | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                              |
|-----------------|---------|-------------------|----------|----------------------------------------------------------------------------------------------------------------|
| Isobutyric acid | 79-31-2 | EEC No. 201-195-7 | >95      | Flam. Liq. 3 (H226)<br>Acute Tox. 4 (H302)<br>Acute Tox. 3 (H311)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318) |

|                           |                  |
|---------------------------|------------------|
| REACH registration number | 01-2119488973-18 |
|---------------------------|------------------|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### 4.3. Indication of any immediate medical attention and special treatment needed

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

#### **Hygiene Measures**

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

Handle in accordance with good industrial hygiene and safety practice.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Corrosives area.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                          | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Isobutyric acid<br>79-31-2 ( >95 ) |                              |                                 |                                | DNEL = 3.75mg/kg<br>bw/day        |

| Component | Acute effects local | Acute effects | Chronic effects local | Chronic effects |
|-----------|---------------------|---------------|-----------------------|-----------------|
|-----------|---------------------|---------------|-----------------------|-----------------|

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

|                                    | (Inhalation) | systemic (Inhalation) | (Inhalation) | systemic (Inhalation)       |
|------------------------------------|--------------|-----------------------|--------------|-----------------------------|
| Isobutyric acid<br>79-31-2 ( >95 ) |              |                       |              | DNEL = 184mg/m <sup>3</sup> |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                          | Fresh water          | Fresh water sediment                | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)               |
|------------------------------------|----------------------|-------------------------------------|--------------------|------------------------------------|----------------------------------|
| Isobutyric acid<br>79-31-2 ( >95 ) | PNEC =<br>0.0451mg/L | PNEC =<br>0.364mg/kg<br>sediment dw | PNEC = 0.451mg/L   | PNEC = 19mg/L                      | PNEC =<br>0.0462mg/kg soil<br>dw |

| Component                          | Marine water         | Marine water sediment                | Marine water Intermittent | Food chain | Air |
|------------------------------------|----------------------|--------------------------------------|---------------------------|------------|-----|
| Isobutyric acid<br>79-31-2 ( >95 ) | PNEC =<br>0.0045mg/L | PNEC =<br>0.0363mg/kg<br>sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

**Environmental exposure controls** Prevent product from entering drains.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                      |                                          |
|------------------------------------------------|--------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                               |                                          |
| <b>Appearance</b>                              | Colorless                            |                                          |
| <b>Odor</b>                                    | Stench                               |                                          |
| <b>Odor Threshold</b>                          | No data available                    |                                          |
| <b>Melting Point/Range</b>                     | -47 °C / -52.6 °F                    |                                          |
| <b>Softening Point</b>                         | No data available                    |                                          |
| <b>Boiling Point/Range</b>                     | 153 - 154 °C / 307.4 - 309.2 °F      | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Flammable                            | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                       | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.6<br><b>Upper</b> 7.3 |                                          |
| <b>Flash Point</b>                             | 59 °C / 138.2 °F                     | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 420 °C / 788 °F                      |                                          |
| <b>Decomposition Temperature</b>               | No data available                    |                                          |
| <b>pH</b>                                      | 2.9                                  | (0.1M)                                   |
| <b>Viscosity</b>                               | 1.1 mPa s at 20 °C                   |                                          |
| <b>Water Solubility</b>                        | 210 g/L (20°C)                       |                                          |
| <b>Solubility in other solvents</b>            | No information available             |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                      |                                          |
| <b>Component</b>                               | <b>log Pow</b>                       |                                          |
| Isobutyric acid                                | 1.1                                  |                                          |
| <b>Vapor Pressure</b>                          | 0.43 mmHg @ 20 °C                    |                                          |
| <b>Density / Specific Gravity</b>              | 0.950                                |                                          |
| <b>Bulk Density</b>                            | Not applicable                       | Liquid                                   |
| <b>Vapor Density</b>                           | 3.04 (Air = 1.0)                     | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)              |                                          |

### 9.2. Other information

|                             |                                        |
|-----------------------------|----------------------------------------|
| <b>Molecular Formula</b>    | C4 H8 O2                               |
| <b>Molecular Weight</b>     | 88.11                                  |
| <b>Explosive Properties</b> | explosive air/vapour mixtures possible |

## SECTION 10: STABILITY AND REACTIVITY

|                         |                                            |
|-------------------------|--------------------------------------------|
| <b>10.1. Reactivity</b> | None known, based on information available |
|-------------------------|--------------------------------------------|

|                                 |                                 |
|---------------------------------|---------------------------------|
| <b>10.2. Chemical stability</b> | Stable under normal conditions. |
|---------------------------------|---------------------------------|

### 10.3. Possibility of hazardous reactions

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>      | None under normal processing.            |

|                                  |                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>10.4. Conditions to avoid</b> | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. |
|----------------------------------|-------------------------------------------------------------------------------------------------------|

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

## 10.5. Incompatible materials

Bases. Strong oxidizing agents. Reducing Agent.

## 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Category 3

Inhalation

Based on available data, the classification criteria are not met

| Component       | LD50 Oral                               | LD50 Dermal          | LC50 Inhalation         |
|-----------------|-----------------------------------------|----------------------|-------------------------|
| Isobutyric acid | 266 mg/kg ( Rat )<br>2230 mg/kg ( Rat ) | 475 mg/kg ( Rabbit ) | > 0.55 mg/L ( Rat ) 4 h |

##### (b) skin corrosion/irritation;

Category 1 B

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

##### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

##### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

##### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

##### (j) aspiration hazard;

Based on available data, the classification criteria are not met

#### Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

### 11.2. Information on other hazards



# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

## Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Do not empty into drains. Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Fish | Water Flea                                  | Freshwater Algae                               |
|-----------------|-----------------|---------------------------------------------|------------------------------------------------|
| Isobutyric acid |                 | EC50: = 51 mg/L, 48h (Daphnia magna Straus) | EC50: = 45 mg/L, 72h (Desmodesmus subspicatus) |

| Component       | Microtox            | M-Factor |
|-----------------|---------------------|----------|
| Isobutyric acid | EC50 = 57 mg/L 17 h |          |

### 12.2. Persistence and degradability

#### Persistence

Persistence is unlikely.

#### Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Isobutyric acid | 1.1     | No data available             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

|                                       |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>              | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.                 |
| <b>Switzerland - Waste Ordinance</b>  | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                 |
|-----------------------------------------|-----------------|
| <b>14.1. UN number</b>                  | UN2529          |
| <b>14.2. UN proper shipping name</b>    | ISOBUTYRIC ACID |
| <b>14.3. Transport hazard class(es)</b> | 3               |
| <b>Subsidiary Hazard Class</b>          | 8               |
| <b>14.4. Packing group</b>              | III             |

### ADR

|                                         |                 |
|-----------------------------------------|-----------------|
| <b>14.1. UN number</b>                  | UN2529          |
| <b>14.2. UN proper shipping name</b>    | ISOBUTYRIC ACID |
| <b>14.3. Transport hazard class(es)</b> | 3               |
| <b>Subsidiary Hazard Class</b>          | 8               |
| <b>14.4. Packing group</b>              | III             |

### IATA

|                                         |                 |
|-----------------------------------------|-----------------|
| <b>14.1. UN number</b>                  | UN2529          |
| <b>14.2. UN proper shipping name</b>    | ISOBUTYRIC ACID |
| <b>14.3. Transport hazard class(es)</b> | 3               |
| <b>Subsidiary Hazard Class</b>          | 8               |
| <b>14.4. Packing group</b>              | III             |

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| <b>14.5. Environmental hazards</b>                                   | No hazards identified            |
| <b>14.6. Special precautions for user</b>                            | No special precautions required. |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods   |

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Isobutyric acid | 79-31-2 | 201-195-7 | -      | -   | X     | X    | KE-24875 | X    | X    |

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

| Component       | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Isobutyric acid | 79-31-2 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Not applicable

| Component       | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Isobutyric acid | 79-31-2 | -                                                                   | -                                                                             | -                                                                                                     |

Seveso III Directive (2012/18/EC)

| Component       | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Isobutyric acid | 79-31-2 | Not applicable                                                                            | Not applicable                                                                           |

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------|---------------------------------------|-------------------------|
| Isobutyric acid | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                        | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Isobutyric acid<br>79-31-2 (>95) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H311 - Toxic in contact with skin  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H226 - Flammable liquid and vapor

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 29-Oct-2010

**Revision Date** 25-Sep-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

Disclaimer

# SAFETY DATA SHEET

Isobutyric acid

Revision Date 25-Sep-2023

---

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**